PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

7 Dec – 13 Dec 2026 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
7 Dec T2026-12-07	Maths Pre-annual mock	___ / 80	Retest: 10 Dec

Maths · Pre-Annual Mock Paper · Answer key

Monday, 7 December 2026 · 80 marks

Test ID: T2026-12-07

Class 9 · Ganita Manjari Ch 1–13 (Ch 14 Surface Area & Volume chhod kar)

Q1 **[1]**

Question: The point (–3, 5) lies in the:

(a) I quadrant (b) II quadrant

(c) III quadrant (d) IV quadrant

Expected answer / marking points: (b) II quadrant

Q2 **[1]**

Question: The distance between the points (2, –4) and (2, 7) is:

(a) 3 (b) 9

(c) 11 (d) 13

Expected answer / marking points: (c) 11 — $|7 - (-4)| = 11$

Q3 **[1]**

Question: The degree of the polynomial $3y^3 - 4y^2 + 5$ is:

(a) 1 (b) 2

(c) 3 (d) 5

Expected answer / marking points: (c) 3

Q4 **[1]**

Question: The slope of the line $y = -2x + 7$ is:

(a) 7 (b) –2

(c) 2 (d) –7

Expected answer / marking points: (b) –2

Q5 **[1]**

Question: Which of the following is irrational?

(a) $\sqrt{16}$ (b) 0.121212...

(c) $\sqrt{7}$ (d) 22/7

Expected answer / marking points: (c) $\sqrt{7}$

Q6 **[1]**

Question: Using a suitable identity, 101^2 =

(a) 10101 (b) 10201

(c) 10001 (d) 10200

Expected answer / marking points: (b) 10201 — $(100 + 1)^2 = 10000 + 200 + 1$

Q7 **[1]**

Question: If $x + 1/x = 5$, then $x^2 + 1/x^2$ =

(a) 25 (b) 27

(c) 23 (d) 10

Expected answer / marking points: (c) 23 — $25 - 2 = 23$

Q8 **Question:** A chord of length 16 cm is drawn in a circle of radius 10 cm. Its distance from the centre is:
[1]

(a) 6 cm (b) 8 cm
(c) 4 cm (d) 12 cm

Expected answer / marking points: (a) 6 cm — $\sqrt{(10^2 - 8^2)} = 6$

Q9 **Question:** The angle subtended by a diameter at any point on the circle is:
[1]

(a) 45° (b) 60°
(c) 90° (d) 180°

Expected answer / marking points: (c) 90°

Q10 **Question:** The area of a sector of radius 7 cm and central angle 90° is:
[1]

(a) 38.5 cm^2 (b) 154 cm^2
(c) 11 cm^2 (d) 77 cm^2

Expected answer / marking points: (a) 38.5 cm^2 — ¼ × 22/7 × 49

Q11 **Question:** A die is thrown once. The probability of getting a number greater than 4 is:
[1]

(a) 1/6 (b) 1/3
(c) 1/2 (d) 2/3

Expected answer / marking points: (b) 1/3 — {5, 6} → 2/6

Q12 **Question:** The common ratio of the GP 5, –10, 20, –40, ... is:
[1]

(a) 2 (b) –2
(c) –5 (d) 1/2

Expected answer / marking points: (b) –2

Q13 **Question:** The 10th term of the AP 2, 7, 12, 17, ... is:
[1]

(a) 45 (b) 47
(c) 52 (d) 50

Expected answer / marking points: (b) 47 — 2 + 9 × 5

Q14 **Question:** The converse of "If a triangle is equilateral, then it is isosceles" is:
[1]

(a) If a triangle is not equilateral, then it is not isosceles.
(b) If a triangle is isosceles, then it is equilateral.
(c) If a triangle is isosceles, then it is not equilateral.
(d) A triangle is equilateral only if it is scalene.

Expected answer / marking points: (b) If a triangle is isosceles, then it is equilateral.

Q15 **Question:** Riya scored 80 in a test of weight 3 and 60 in a test of weight 1. Her weighted average is:
[1]

(a) 70 (b) 75
(c) 72 (d) 65

Expected answer / marking points: (b) 75 — (240 + 60) ÷ 4

Q16 **[1]** **Question:** The line $x + 2y = 6$ cuts the y-axis at:

(a) (6, 0) (b) (0, 6)

(c) (0, 3) (d) (3, 0)

Expected answer / marking points: (c) (0, 3)

Q17 **[1]** **Question:** An algorithm checks every number j from 1 to n to see whether j divides n. For n = 20, how many numbers does it check?

(a) 10

(b) 20

(c) 6

(d) 4

Expected answer / marking points: (b) 20

Q18 **[1]** **Question:** In a parallelogram ABCD, $\angle A = 70^\circ$. Then $\angle C$ =

(a) 110° (b) 70°

(c) 20° (d) 140°

Expected answer / marking points: (b) 70°

Q19 **[1]** **Question: A:** $(x + y)^2 = x^2 + y^2$ is an identity.
R: An identity is an equation that is true for all values of the variables.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d) — A is false (it fails for $x = y = 1$); R is the correct definition.

Q20 **[1]** **Question: A:** The centroid of a triangle divides each median in the ratio 2 : 1.
R: The three medians of a triangle are concurrent.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (b) — both true; concurrency alone does not explain the 2 : 1 ratio.

Q21 **[2]** **Question:** Points A(–3, 4) and B(5, 4) are given. Find the length AB and the point on AB that is equidistant from A and B.

Expected answer / marking points: Same y-coordinate, so $AB = |5 - (-3)| = 8$ units [1]. Equidistant point = midpoint = $((-3 + 5)/2, 4) = (1, 4)$ [1].

Q22 **[2]** **Question:** (i) Factorise $4x^2 - 12x + 9$. (ii) Evaluate 98×102 using a suitable identity.

Expected answer / marking points: (i) $(2x)^2 - 2(2x)(3) + 3^2 = (2x - 3)^2$ [1]. (ii) $(100 - 2)(100 + 2) = 10000 - 4 = 9996$ [1].

Q23 **[2]** **Question:** Find two rational numbers between 1/3 and 1/2.

Expected answer / marking points: 1/3 = 4/12, 1/2 = 6/12, so 5/12 is between them [1]; also 1/3 = 8/24, 1/2 = 12/24 gives 9/24 = 3/8 (or 10/24, 11/24) [1]. Any two correct.

Q24 **[2]** **Question:** A bag has 5 red, 3 green and 2 yellow balls. One ball is drawn at random. Find (i) P(green) (ii) P(not yellow).

Expected answer / marking points: Total = 10. (i) 3/10 [1] (ii) 8/10 = 4/5 [1].

Q25 **[2]** **Question:** Write the converse of: "If two angles are vertically opposite, then they are equal." Is the converse true? Justify.

Expected answer / marking points: Converse: If two angles are equal, then they are vertically opposite [1]. False — e.g., two angles of an equilateral triangle are equal (60° each) but are not vertically opposite [1].

Q26 **[3]** **Question:** Prove that $\sqrt{2}$ is irrational.

Expected answer / marking points: Assume $\sqrt{2} = p/q$ in lowest terms ($q \neq 0$) [0.5]. Then $p^2 = 2q^2$, so p^2 is even $\Rightarrow$ p is even, $p = 2k$ [1]. Then $4k^2 = 2q^2 \Rightarrow q^2 = 2k^2 \Rightarrow$ q is even [1]. Both p and q even contradicts lowest terms, so $\sqrt{2}$ is irrational [0.5].

Q27 **[3]** **Question:** Find the area of a triangle whose sides are 13 cm, 14 cm and 15 cm using Heron's formula.

Expected answer / marking points: $s = (13 + 14 + 15)/2 = 21$ [1]. Area = $\sqrt{(21 \times 8 \times 7 \times 6)}$ [1] = $\sqrt{7056} = 84$ cm^2 [1].

Q28 **[3]** **Question:** Rahul saves ₹50 in week 1, ₹65 in week 2, ₹80 in week 3, and so on. (i) How much does he save in week 20? (ii) In which week will he first save more than ₹500?

Expected answer / marking points: AP with $a = 50$, $d = 15$. (i) $t_{20} = 50 + 19 \times 15 =$ ₹335 [1.5]. (ii) $50 + (n - 1)15 > 500 \Rightarrow n - 1 > 30 \Rightarrow n = 32$ ($t_{31} = 500$ is not more than 500; $t_{32} = 515$) [1.5].

Q29 **[3]** **Question:** AB and CD are parallel chords of a circle of radius 13 cm, on opposite sides of the centre. AB = 10 cm and CD = 24 cm. Find the distance between the chords.

Expected answer / marking points: Perpendicular from centre bisects a chord. Distance of AB = $\sqrt{(13^2 - 5^2)} = 12$ cm [1]; of CD = $\sqrt{(13^2 - 12^2)} = 5$ cm [1]. Opposite sides $\Rightarrow$ distance = 12 + 5 = 17 cm [1].

Q30 **[3]** **Question:** Class A has 30 students with average marks 72, and Class B has 20 students with average 82. (i) Find the combined average. (ii) A new student scoring 95 joins Class B. Find the new average of Class B.

Expected answer / marking points: (i) $(30 \times 72 + 20 \times 82)/50 = (2160 + 1640)/50 = 76$ [1.5]. (ii) $(1640 + 95)/21 = 1735/21 \approx 82.62$ [1.5].

Q31 **[3]** **Question:** Solve by the method of elimination: $2x + 3y = 13$ and $3x - 2y = 0$.

Expected answer / marking points: Multiply: $4x + 6y = 26$ and $9x - 6y = 0$ [1]; add: $13x = 26 \Rightarrow x = 2$ [1]; then $3y = 13 - 4 \Rightarrow y = 3$ [0.5]. Check: $6 - 6 = 0$ ✓ [0.5].

Q32 [5] **Question:** Draw the graphs of $y = 2x - 1$ and $x + y = 5$ on the same axes. (i) Find the point where they intersect. (ii) Find the points where each line cuts the y-axis. (iii) Find the area of the triangle formed by the two lines and the y-axis.

ATHVA / OR

The cost of 2 pens and 3 notebooks is ₹130, and the cost of 3 pens and 2 notebooks is ₹120. Form the pair of linear equations and solve them by substitution to find the cost of one pen and one notebook.

Expected answer / marking points: Tables of at least 3 points each, e.g., $y = 2x - 1$: (0, –1), (1, 1), (2, 3); $x + y = 5$: (0, 5), (2, 3), (5, 0) [1]; correct graph [1.5]. (i) (2, 3) [1]. (ii) (0, –1) and (0, 5) [0.5]. (iii) Base on y-axis = 6, height = x-coordinate of (2, 3) = 2; area = ½ × 6 × 2 = 6 sq units [1].
OR: $2p + 3n = 130$, $3p + 2n = 120$ [1]. From the first, $p = (130 - 3n)/2$ [1]; substitute: $3(130 - 3n)/2 + 2n = 120 \Rightarrow 390 - 9n + 4n = 240 \Rightarrow n = 30$ [1.5]; $p = (130 - 90)/2 = 20$ [1]. Pen ₹20, notebook ₹30; check $60 + 60 = 120$ ✓ [0.5].

Q33 [5] **Question:** (i) Factorise $x^3 - 27$. (ii) Without calculating the cubes, find $(-12)^3 + 7^3 + 5^3$. (iii) Expand $(2x + 3y)^3$.

Expected answer / marking points: (i) $(x - 3)(x^2 + 3x + 9)$ [1]. (ii) $-12 + 7 + 5 = 0$, so $a^3 + b^3 + c^3 = 3abc = 3 \times (-12) \times 7 \times 5 = -1260$ [2]. (iii) $8x^3 + 36x^2y + 54xy^2 + 27y^3$ [2].

Q34 [5] **Question:** Prove that the quadrilateral formed by joining the midpoints of the sides of any quadrilateral (in order) is a parallelogram.

ATHVA / OR

Prove that the diagonals of a parallelogram bisect each other.

Expected answer / marking points: Figure: ABCD with midpoints P, Q, R, S of AB, BC, CD, DA; draw diagonal AC [1]. In △ABC, PQ ∥ AC and PQ = ½AC (Midpoint Theorem) [1.5]. In △ADC, SR ∥ AC and SR = ½AC [1.5]. So PQ ∥ SR and PQ = SR — one pair of opposite sides equal and parallel ⇒ PQRS is a parallelogram [1].
OR: Figure: parallelogram ABCD, diagonals meet at O [1]. In △AOB and △COD: AB = CD (opposite sides) [1]; ∠OAB = ∠OCD and ∠OBA = ∠ODC (alternate angles, AB ∥ CD) [1.5]; so △AOB ≅ △COD (ASA) [1] ⇒ OA = OC and OB = OD [0.5].

Q35 [5] **Question:** A spinner has three equal parts coloured Red, Blue and Green. It is spun twice. (i) Draw a tree diagram and write the sample space. Find the probability of getting (ii) the same colour both times (iii) at least one Red (iv) no Green.

Expected answer / marking points: (i) Tree with 3 × 3 branches [1]; S = {RR, RB, RG, BR, BB, BG, GR, GB, GG}, 9 outcomes [1]. (ii) RR, BB, GG ⇒ 3/9 = 1/3 [1]. (iii) 5 outcomes have R ⇒ 5/9 [1]. (iv) RR, RB, BR, BB ⇒ 4/9 [1].

Q36 [4] **Question:**

> A park is drawn on a grid with the main gate at the origin O(0, 0). The fountain is at F(4, 3), a bench at B(–2, 3) and a lamp post at L(4, –5). 1 unit = 10 m.

(i) In which quadrant is the lamp post? [1]
(ii) Find the distance FB. [1]
(iii) Find the area of triangle FBL in square units and in m^2. [2]

Expected answer / marking points: (i) IV quadrant [1]. (ii) Same y ⇒ FB = |4 – (–2)| = 6 units (60 m) [1]. (iii) FL = |3 – (–5)| = 8 units; FB is horizontal and FL vertical, so ∠F = 90° [0.5]; area = ½ × 6 × 8 = 24 sq units [1] = 24 × 100 = 2400 m^2 [0.5].

Q37 **Question:**

[4]

> A school gives final marks as a weighted average: Unit test 20%, Half-yearly 30%, Annual 50%. Riya scored 80 in the unit test, 70 in the half-yearly and 90 in the annual exam.

(i) Find Riya's final (weighted) score. [1]
(ii) Find the simple average of her three scores. [1]
(iii) Aman scored 80 and 70 in the first two exams. What must he score in the annual exam to get a final score of 85? [2]

Expected answer / marking points: (i) 0.2 × 80 + 0.3 × 70 + 0.5 × 90 = 16 + 21 + 45 = 82 [1]. (ii) (80 + 70 + 90)/3 = 80 [1]. (iii) 16 + 21 + 0.5a = 85 [1] ⇒ 0.5a = 48 ⇒ a = 96 [1].

Q38 **Question:**

[4]

> In a Sierpiński triangle, we start (stage 0) with one shaded triangle. At every stage, each shaded triangle is replaced by 3 smaller shaded triangles, and the shaded area becomes ¾ of the shaded area of the previous stage.

(i) How many shaded triangles are there at stage 3? [1]
(ii) Write the number of shaded triangles at stage n. What kind of progression is this? [1]
(iii) If the shaded area at stage 0 is 64 cm^2, find the shaded area at stage 3. [2]

Expected answer / marking points: (i) 1, 3, 9, 27 ⇒ 27 [1]. (ii) 3^n; a GP with common ratio 3 [1]. (iii) Areas form a GP with r = ¾: 64 × $(¾)^3$ [1] = 64 × 27/64 = 27 cm^2 [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: